CE205: Databases and Information Retrieval

Lecture 1 Activities

Problem

Consider the relation HIGHWAY with sample data (*the underlined attributes represent the primary key, and assume that there is no space in the attributes' name*).

ProjectID	**Project Name**	**CountyID**	**County Name**	**Project Manager ID**	**Project Manager Name**	**ProjectMiles WithinCounty**
1	Road X	1	Wilson	M1	Bob	10.00
1	Road X	2	Ottawa	M1	Bob	17.00
1	Road X	3	Davis	M1	Bob	12.00
2	Road Y	3	Davis	M2	Sue	23.00
3	Bridge A	1	Wilson	M3	Lee	0.50
3	Bridge A	2	Ottawa	M3	Lee	0.30
4	Tunnel Q	2	Ottawa	M1	Bob	2.00
5	Road W	4	Pony	M4	Rob	23.00

- *The HIGHWAY Project Table captures the data about projects and their length (in miles).*
- *Each project has a unique Project ID and Project Name.*
- *Each county has a unique County ID and County Name.*
- *Each project manager has a unique Project Manager ID and Project Manager Name.*
- *Each project has one project manager.*
- *A project manager can manage several projects.*
- *A project can span across several counties.*
- *This table records the length of the project in a county in the ProjectMilesWithinCounty column*

(a) Using the relation HIGHWAY, provide examples that illustrate insertion anomaly, delete anomaly, and modification anomaly.

(b) Determine all possible functional dependencies in the relation HIGHWAY.

ProjectID	**Project Name**	**CountyID**	**County Name**	**Project Manager ID**	**Project Manager Name**	**ProjectMiles WithinCounty**